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United States Patent Application Publication

20250286082

Kind Code

A1

Publication Date

September 11, 2025

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Electrodes For Microbial Fuel Cells, Microbial Fuel Cell, And Microbial Fuel Cell System

Abstract

A electrode for MFCs includes: a non-conductive fiber; a conductive material having conductivity; and a binder configured to bind the conductive material to the fiber.

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Family ID: 96949866

Appl. No.: 19/070890

Filed: March 05, 2025

Foreign Application Priority Data

JP	2024-033663	Mar. 06, 2024
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Publication Classification

Int. Cl.: H01M4/86 (20060101); H01M8/10 (20160101); H01M8/1004 (20160101); H01M8/16 (20060101)

U.S. Cl.:

CPC H01M4/8657 (20130101); H01M4/8668 (20130101); H01M4/8673 (20130101); H01M8/1004 (20130101); H01M8/16 (20130101); H01M2008/1095 (20130101)

Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-033663, filed Mar. 6, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to electrodes for microbial fuel cells (MFCs), a microbial fuel cell (MFC), and a MFC system.

2. Related Art

[0003] MFCs that utilize metabolic capabilities of microorganisms are known. The MFC described in JP-A-2017-117584 includes a negative electrode, a hydrogen ion transfer layer, and a positive electrode. The negative electrode is formed of a conductive sheet having conductivity. The conductive sheet may be a metal plate or a graphite sheet having a through hole. The positive electrode includes a graphite sheet, a filter layer, and a water-repellent layer.

[0004] The graphite sheet used in the negative or positive electrode has insufficient electrical properties as an electrode.

SUMMARY

[0005] A electrodes for MFCs according to the present disclosure includes: a non-conductive fiber; a conductive material having conductivity; and a binder configured to bind the conductive material to the fiber.

[0006] A MFC according to the present disclosure is a MFC in contact with a liquid containing an organic substance and an electrochemically active bacterium, including: an anode electrode containing a non-conductive fiber, a conductive material having conductivity, and a binder configured to bind the conductive material to the fiber; and a cathode electrode having conductivity.

[0007] A MFC system according to the present disclosure includes: a MFC including an anode electrode to be in contact with a liquid containing an organic substance and an electrochemically active bacterium, a cathode electrode to which oxygen is supplied, and a proton exchange membrane (PEM) disposed between the anode electrode and the cathode electrode; and a resistor coupled to the anode electrode and the cathode electrode. The anode electrode contains a non-conductive fiber, a conductive material having conductivity, and a binder configured to bind the conductive material to the fiber.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram showing a schematic configuration of a MFC system.

[0009] FIG. 2 is a diagram showing an example of a configuration of an anode electrode.

[0010] FIG. 3 is a diagram showing an example of the configuration of the anode electrode.

[0011] FIG. 4 is a diagram showing an example of the configuration of the anode electrode.

[0012] FIG. 5 is a diagram showing a schematic configuration of a cathode electrode.

[0013] FIG. 6 is a diagram showing an example of a configuration of the cathode electrode.

[0014] FIG. 7 is a diagram showing a configuration of a MFC.

[0015] FIG. 8 is a diagram showing an evaluation result of the MFC.

[0016] FIG. 9 is a diagram showing the evaluation result of the MFC.

[0017] FIG. 10 is a diagram showing the evaluation result of the MFC.

[0018] FIG. 11 is a diagram showing the evaluation result of the MFC.

[0019] FIG. 12 is a diagram showing the evaluation result of the MFC.

DESCRIPTION OF EMBODIMENTS

[0020] FIG. 1 shows a schematic configuration of a MFC system **100**. The MFC system **100**

decomposes organic substances contained in an organic wastewater L using microorganisms. The MFC system **100** produces energy, useful substances, and the like by electrochemically controlling metabolism of the microorganisms. The MFC system **100** includes a MFC **10**, an accommodation container **20**, and an external circuit **30**. The accommodation container **20** accommodates the organic wastewater L containing organic substances and microorganisms.

[0021] The MFC **10** converts the organic substance, which is a fuel, into electric energy using the microorganisms. The MFC **10** is immersed in the organic wastewater L accommodated in the accommodation container **20**. The MFC **10** includes an anode electrode **11**, a PEM **13**, a cathode electrode **15**, and a shielding member **17**.

[0022] The anode electrode **11** collects electrons generated when the organic substance in the organic wastewater L is oxidized and decomposed by the microorganism. The electrons collected by the anode electrode **11** move to the cathode electrode **15** via the external circuit **30**. The anode electrode **11** is in contact with the organic wastewater L accommodated in the accommodation container **20**. On the anode electrode **11**, the microorganism contained in the organic wastewater L decomposes the organic substance to generate electrons and hydrogen ions. Electrons generated by the microorganisms are collected by the anode electrode **11**. The anode electrode **11** includes fiber bodies **111**, conductive materials **113**, and binding materials **115**. A detailed configuration of the anode electrode **11** will be described later. The anode electrode **11** corresponds to an example of a electrodes for MFCs.

[0023] The anode electrode **11** is formed in a hollow cylindrical shape. The anode electrode **11** may have a columnar shape or a quadrangular prism shape. The anode electrode **11** may be constituted by one member or may be constituted by combining a plurality of members. An outer peripheral surface of the anode electrode **11** is in contact with the organic wastewater L and carries the microorganism in the organic wastewater L.

[0024] The PEM **13** allows hydrogen ions generated when a microorganism decomposes an organic substance to pass therethrough. The PEM **13** has a function of moving the hydrogen ions from the anode electrode **11** to the cathode electrode **15**.

[0025] The PEM **13** is disposed on an inner peripheral surface of the anode electrode **11**. The PEM **13** is formed in a hollow cylindrical shape. An outer peripheral surface of the PEM **13** is in contact with the inner peripheral surface of the anode electrode **11**. The inner peripheral surface of the PEM **13** is in contact with the cathode electrode **15**. The PEM **13** is disposed between the anode electrode **11** and the cathode electrode **15**, and separates the anode electrode **11** from the cathode electrode **15**.

[0026] The PEM **13** is made of, for example, an ion exchange resin. The ion exchange resin includes NAFION manufactured by DuPont Kabushiki Kaisha, FLEMION manufactured by AGC Inc., and SELEMION manufactured by AGC Inc. NAFION, FLEMION, and SELEMION are registered trademarks.

[0027] The PEM **13** may be formed of a porous membrane having pores through which hydrogen ions permeate. The PEM **13** includes a porous sheet, a woven fabric sheet, and a nonwoven fabric sheet. The PEM **13** is preferably a sheet material containing a non-conductive fiber material, wood powder particles, and an adhesive for bonding the fiber material to the wood powder particles. The fiber material may be the same as or different from the fiber body **111** included in the anode electrode **11**. The fiber material is preferably the same material as the fiber body **111**. Since the fiber material is the same material as the fiber body **111**, adhesion between the anode electrode **11** and the PEM **13** is improved.

[0028] The cathode electrode **15** consumes electrons moving through the external circuit **30** in a reduction reaction of an oxidant. The electrons flow in the external circuit **30** according to a gradient of a potential generated in the anode electrode **11** and an oxidation-reduction potential of a chemical reaction generated in the cathode electrode **15**. The cathode electrode **15** uses oxygen in an atmosphere as an oxidant. Oxygen moves in the cathode electrode **15**. Oxygen reacts with

hydrogen ions moving in the PEM **13**.

[0029] The cathode electrode **15** is formed in a hollow cylindrical shape. An ion exchange surface **15a**, which is an outer peripheral surface of the cathode electrode **15**, is in contact with the inner peripheral surface of the PEM **13** directly or via another member. An oxygen-permeable surface **15b**, which is an inner peripheral surface of the cathode electrode **15**, is in contact with the atmosphere in an atmospheric space AS. The atmospheric space AS is formed in an inner peripheral portion of the cathode electrode **15**. The oxygen-permeable surface **15b** allows oxygen in the atmosphere to permeate through the cathode electrode **15**. Oxygen is supplied to the cathode electrode **15**. The configuration of the cathode electrode **15** will be described later.

[0030] The shielding member **17** is disposed at one end of the oxygen-permeable surface **15b**. The shielding member **17** prevents the organic wastewater L from flowing into the atmospheric space AS. The form of the shielding member **17** is not limited as long as the organic wastewater L is prevented from flowing into the atmospheric space AS.

[0031] The accommodation container **20** accommodates the organic wastewater L. The accommodation container **20** may be coupled to an inflow path and an outflow path. The inflow path is a path through which the organic wastewater L flows into the accommodation container **20**. The outflow path is a path through which the organic wastewater L flows out from the accommodation container **20**. The inflow path and the outflow path are not shown. The MFC system **100** shown in FIG. **1** includes the accommodation container **20**, and is not limited thereto. The accommodation container **20** may not be provided. When the accommodation container **20** is not provided, the MFC **10** is disposed in a lake, a river, or the like.

[0032] The organic wastewater L contains the organic substance and the microorganism. The organic wastewater L is, for example, water collected from lakes or rivers. The organic wastewater L may be water or living wastewater containing organic mud collected from lakes or the like. The organic substance in the organic wastewater L is consumed as a fuel of the microorganism.

[0033] The microorganisms are electrochemically active bacteria that decompose the organic substance in the organic wastewater L. The electrochemically active bacteria include *Geobacter* bacteria, *Shewanella* bacteria, *Aeromonas* bacteria, *Geothrix* bacteria, and *Saccharomyces* bacteria. The microorganisms may be contained in the organic wastewater L in advance, or may be added to the organic wastewater L. The microorganisms may be supported by the anode electrode **11** with a biofilm or the like.

[0034] The external circuit **30** is an electric circuit that moves electrons from the anode electrode **11** to the cathode electrode **15**. The external circuit **30** is electrically coupled to the anode electrode **11** and the cathode electrode **15**. The external circuit **30** includes a resistor **31**. The resistor **31** is coupled to the anode electrode **11** and the cathode electrode **15**. The resistor **31** may be a variable resistor for which a resistance value is switched. The resistor **31** adjusts an amount of current flowing through the external circuit **30**.

[0035] In the MFC **10** shown in FIG. **1**, the anode electrode **11**, the PEM **13**, and the cathode electrode **15** are integrally formed, and the present disclosure is not limited thereto. The anode electrode **11**, the PEM **13**, and the cathode electrode **15** may be separately formed. The anode electrode **11**, the PEM **13**, and the cathode electrode **15** are each immersed in the organic wastewater L.

[0036] FIG. **2** shows an example of the configuration of the anode electrode **11**. FIG. **2** schematically shows an enlarged configuration of a part of the anode electrode **11**. The anode electrode **11** shown in FIG. **2** includes the fiber body **111**, a conductive powder **113a** as an example of the conductive material **113**, and the binding material **115**.

[0037] The fiber body **111** is made of a non-conductive material. The fiber body **111** carries the conductive powder **113a**. Since the anode electrode **11** includes the fiber bodies **111**, the number of holes in the anode electrode **11** increases. As the number of holes increases, a surface area of the anode electrode **11** increases. As the surface area of the anode electrode **11** increases, the number of

sites where electrons are transferred by a microorganism increases. The fiber body **111** is, for example, a defibrated fiber defibrated by a dry fiber technology. The defibrated fibers include cellulose fibers obtained by defibrating waste paper, plant fibers obtained by defibrating plants such as *eucalyptus*, and the like. The fiber body **111** has a fiber diameter of 20 μm to 100 μm and a fiber length of 0.5 mm to 100 mm. The fiber bodies **111** are bonded to each other by the binding material **115**. The fiber body **111** is preferably made of an organic material. When the fiber body **111** is eluted into the organic wastewater L due to deterioration or the like, the eluted fiber body **111** is consumed as a fuel of a microorganism. The fiber body **111** corresponds to an example of a fiber. [0038] The conductive powder **113a** has conductivity. The conductive powder **113a** contributes to the conductivity of the anode electrode **11**. The conductive powder **113a** is bonded to a surface of the fiber body **111** by the binding material **115**. A plurality of conductive powders **113a** are in contact with each other and bonded onto the fiber body **111**. The plurality of conductive powders **113a** are disposed in contact with each other, thereby imparting the conductivity to the fiber body **111**. The conductive powder **113a** is disposed in a single layer or a multiple layer on the fiber body **111**. The conductive powder **113a** is preferably disposed in a multiple layer on the fiber body **111**. When the conductive powder **113a** on an outermost surface falls off as a result of the conductive powder **113a** being disposed in the multiple layer, fresh conductive powder **113a** is deposited on the surface. Since the fresh conductive powder **113a** is deposited on the surface, durability of the MFC **10** is improved.

[0039] The conductive powder **113a** is formed of graphite particles, carbon black particles, or the like. The conductive powder **113a** is a powder having an average particle diameter of 0.1 μm to 500 μm on a volume basis. The conductive powder **113a** is formed of one or a plurality of types of powders. The material of the conductive powder **113a** is not limited as long as the conductive powder **113a** has a property of imparting conductivity to the fiber body **111**.

[0040] A content of the conductive powder **113a** is preferably 50 mass % or more and 80 mass % or less in terms of a mass ratio of the anode electrode **11**. When the content of the conductive powder **113a** is less than 50 mass %, an electric resistance of the anode electrode **11** increases, and a function as an electrode deteriorates. When the content of the conductive powder **113a** is more than 80 mass %, holes in the anode electrode **11** are reduced.

[0041] The binding material **115** binds the conductive powder **113a** to the fiber body **111**. The conductive powder **113a** is bonded to the surface of the fiber body **111** by the binding material **115**. The binding material **115** binds the plurality of fiber bodies **111** to each other. The plurality of fiber bodies **111** are bonded to each other by the binding material **115**. The binding material **115** binds the plurality of conductive powders **113a** to each other. Due to the binding material **115**, the plurality of conductive powders **113a** are formed in one or more layers on the fiber body **111**.

[0042] The material of the binding material **115** is not limited as long as the fiber body **111** and the conductive powder **113a** can be bonded. The binding material **115** may be a solid material or a liquid material at a room temperature. The binding material **115** preferably contains at least one of calcium carbonate, a lignin powder, and a starch powder. The binding material **115** contains at least one of calcium carbonate, a lignin powder (average particle diameter of 26 μm to 70 μm on a volume basis), and a starch powder (average particle diameter of 1 μm to 30 μm on a volume basis), thereby reducing an environmental impact when producing the anode electrode **11**. In addition, a binding force when bonding the plurality of conductive powders **113a** is made to be relatively small. When the conductive powder **113a** is formed of multiple layers, the outermost conductive powder **113a** is likely to fall off. When the outermost conductive powder **113a** falls off, the fresh conductive powder **113a** is exposed. The binding material **115** may be a waxy substance produced from resource organisms such as scale insects. The binding material **115** corresponds to an example of a binder.

[0043] The content of the binding material **115** is preferably 5 mass % or more and 30 mass % or less in terms of the mass ratio of the anode electrode **11**. When the content of the binding material

115 is less than 5 mass %, the binding force of the conductive powder **113a** to be bonded onto the fiber body **111** decreases. It is difficult to maintain a shape of the anode electrode **11**. When the content of the binding material **115** is more than 30 mass %, the amount of the binding material **115** disposed between the plurality of conductive powders **113a** increases, and the electric resistance increases. Electrical properties of the anode electrode **11** deteriorate.

[0044] The anode electrode **11** includes the non-conductive fiber bodies **111**, the conductive materials **113** having conductivity, and the binding materials **115** that bonds the conductive materials **113** to the fiber bodies **111**.

[0045] Since the anode electrode **11** includes the fiber bodies **111**, the conductive materials **113**, and the binding materials **115**, the number of sites where electrons are moved by the microorganism increases. Electrical properties of the anode electrode **11** are improved.

[0046] The binding material **115** preferably contains at least one of calcium carbonate, a lignin powder, and a starch powder.

[0047] When the binding material **115** contains at least one of the calcium carbonate, the lignin powder, and the starch powder, an environmental load when the anode electrode **11** is generated is reduced.

[0048] The content of the fiber body **111** in the anode electrode **11** is a remaining amount excluding the conductive powder **113a** and the binding material **115**. The anode electrode **11** may contain an additive other than the fiber bodies **111**, the conductive powders **113a**, and the binding materials **115**. When the anode electrode **11** includes an additive other than the fiber bodies **111**, the conductive powders **113a**, and the binding materials **115**, the content of the fiber bodies **111** in the anode electrode **11** is a remaining amount excluding the conductive powders **113a**, the binding materials **115**, and the additive.

[0049] The anode electrode **11** is produced using the fiber bodies **111**, the conductive powders **113a**, and the binding materials **115**. A producer mixes the fiber bodies **111** and the binding materials **115** at a predetermined mass ratio to produce a mixture. In the mixture, the binding materials **115** adhere to the surfaces of the fiber bodies **111**. After producing the mixture, the producer adds the conductive powder **113a** to the mixture and mixes them again. The producer generates an electrode mixture including the fiber body **111**, the conductive powder **113a**, and the binding material **115**. The producer applies heat and pressure to the electrode mixture in a predetermined mold to produce a solid body. The applied pressure is in a range of 1 kN/mm^{sup.2} or more and 20 kN/mm^{sup.2} or less. A heating temperature is in a range of 80° C. or higher and 150° C. or lower. The applied pressure and the heating temperature are appropriately adjusted according to addition amounts of the conductive powder **113a** and the binding material **115**. The producer processes the solid body into a desired shape to produce the anode electrode **11**. The anode electrode **11** is a compact including holes.

[0050] FIG. 3 shows an example of the configuration of the anode electrode **11**. FIG. 3 schematically shows an enlarged configuration of a part of the anode electrode **11**. The anode electrode **11** shown in FIG. 3 includes the fiber bodies **111**, the conductive powders **113a**, the binding materials **115**, and zero-valent iron particles **117**. The anode electrode **11** shown in FIG. 3 has a configuration same as that of the anode electrode **11** shown in FIG. 2 except that the anode electrode contains zero-valent iron particles **117**.

[0051] The anode electrode **11** shown in FIG. 3 contains the zero-valent iron particles **117**. The zero-valent iron particles **117** have conductivity. The zero-valent iron particles **117** are iron powders. The zero-valent iron particles **117** are added when the conductive powder **113a** are mixed into the mixture of the fiber bodies **111** and the binding materials **115**. The zero-valent iron particles **117** are powders having an average particle diameter of 0.01 to 100 μm on a volume basis. An amount of the zero-valent iron particles **117** added is 1 mass % or more and 10 mass % or less in terms of the mass ratio of the anode electrode **11**. The addition of the zero-valent iron particles **117** reduces the electric resistance of the anode electrode **11** and improves the electrical properties.

The zero-valent iron particles **117** correspond to an example of metal particles. The anode electrode **11** shown in FIG. **3** includes the zero-valent iron particles **117**, and is not limited thereto. The material is not limited as long as the material reduces the electric resistance of the anode electrode **11**.

[0052] The anode electrode **11** preferably contains the zero-valent iron particles **117**.

[0053] Since the anode electrode **11** contains the zero-valent iron particles **117**, the electric resistance of the anode electrode **11** further deteriorates.

[0054] FIG. **4** shows an example of the configuration of the anode electrode **11**. FIG. **4** schematically shows an enlarged configuration of a part of the anode electrode **11**. The anode electrode **11** shown in FIG. **4** includes the fiber bodies **111**, a conductive fiber **113b** as an example of the conductive material **113**, and the binding materials **115**. The anode electrode **11** shown in FIG. **4** has a configuration same as that of the anode electrode **11** shown in FIG. **2** except that the conductive fiber **113b** is included instead of the conductive powder **113a**. FIG. **4** shows the anode electrode **11** from which the binding material **115** is omitted.

[0055] The conductive fiber **113b** has conductivity. The conductive fiber **113b** contributes to the conductivity of the anode electrode **11**. The conductive fiber **113b** is bonded to the surface of the fiber body **111** by the binding material **115**. A plurality of conductive fibers **113b** are in contact with each other and bonded onto the fiber body **111**. The plurality of conductive fibers **113b** are disposed in contact with each other, thereby imparting the conductivity to the fiber body **111**.

[0056] The conductive fiber **113b** is made of a carbon fiber. The carbon fiber is produced by cutting a polyacrylonitrile (PAN)-based carbon fiber and a pitch-based carbon fiber. The PAN-based carbon fiber is obtained by carbonizing a PAN precursor. The pitch-based carbon fiber is obtained by carbonizing a pitch fiber made of coal tar or heavy petroleum fractions as a raw material. The conductive fibers **113b** have an average fiber diameter of 5 μm to 30 μm and an average fiber length of 0.05 mm to 0.1 mm. A fiber diameter and a fiber length of the conductive fiber **113b** are smaller than the fiber diameter and the fiber length of the fiber body **111**. The conductive fiber **113b** may be made of one or a plurality of types of carbon fibers. The material of the conductive fiber **113b** is not limited as long as the conductive fiber **113b** has a property of imparting the conductivity to the fiber body **111**.

[0057] The anode electrode **11** is produced using the fiber bodies **111**, the conductive fibers **113b**, and the binding materials **115**. A producer mixes the fiber bodies **111** and the binding materials **115** at a predetermined mass ratio to produce a mixture. In the mixture, the binding materials **115** adhere to the surfaces of the fiber bodies **111**. After producing the mixture, the producer adds the conductive fiber **113b** to the mixture and mixes them again. The producer generates an electrode mixture including the fiber body **111**, the conductive fiber **113b**, and the binding material **115**. The producer applies heat and pressure to the electrode mixture in a predetermined mold to produce a solid body. The applied pressure is in a range of 1 kN/mm.² or more and 20 kN/mm.² or less. A heating temperature is in a range of 80° C. or higher and 150° C. or lower. The applied pressure and the heating temperature are appropriately adjusted according to addition amounts of the conductive fiber **113b** and the binding material **115**. The producer processes the solid body into a desired shape to produce the anode electrode **11**. The anode electrode **11** is a compact including holes.

[0058] The anode electrode **11** shown in FIG. **4** may contain the zero-valent iron particles **117**. The zero-valent iron particles **117** are added when the conductive fiber **113b** are mixed into the mixture of the fiber bodies **111** and the binding materials **115**.

[0059] The anode electrode **11** shown in FIG. **4** includes carbon fibers as the conductive fibers **113b**, and is not limited thereto. The anode electrode **11** shown in FIG. **4** may contain the conductive powder **113a** such as graphite particles. The anode electrode **11** preferably includes at least one of the conductive fiber **113b** and the conductive powder **113a**.

[0060] Since the anode electrode **11** contains at least one of the conductive fiber **113b** and the

conductive powder **113a**, the electric resistance of the anode electrode **11** deteriorates.

[0061] FIG. **5** shows a schematic configuration of the cathode electrode **15**. The cathode electrode **15** has the ion exchange surface **15a** and the oxygen-permeable surface **15b**. The cathode electrode **15** includes a water-repellent layer **18** and a filter layer **19**.

[0062] The cathode electrode **15** has conductivity and oxygen permeability. The cathode electrode **15** is formed of a carbon fiber sheet or the like. The cathode electrode **15** may include non-conductive fibers **151**, carbon-based materials **153**, and fixing materials **155**.

[0063] The ion exchange surface **15a** is a surface facing the PEM **13**. The ion exchange surface **15a** is in contact with the PEM **13** directly or via the filter layer **19**. Hydrogen ions react with oxygen on the ion exchange surface **15a** or in the vicinity of the ion exchange surface **15a**.

[0064] The oxygen-permeable surface **15b** is a surface that comes into contact with an atmosphere directly or via the water-repellent layer **18**. The oxygen-permeable surface **15b** allows oxygen in the atmosphere to permeate into the cathode electrode **15**.

[0065] The water-repellent layer **18** is disposed on the oxygen-permeable surface **15b** of the cathode electrode **15**. The water-repellent layer **18** has oxygen permeability and water repellency. The water-repellent layer **18** allows oxygen in the atmosphere to permeate through the cathode electrode **15**. The water-repellent layer **18** prevents the organic wastewater L from adhering to the cathode electrode **15**. The water-repellent layer **18** is made of a nonwoven fabric such as polyethylene or polypropylene, a film such as polytetrafluoroethylene (PTFE), a composite material film in which a polyurethane polymer or the like is combined, beeswax, or the like. Since the water-repellent layer **18** is made of beeswax as an example, the environmental load at the time of producing the cathode electrode **15** is reduced.

[0066] The cathode electrode **15** preferably includes the water-repellent layer **18** disposed on the oxygen-permeable surface **15b** different from the ion exchange surface **15a** in contact with the PEM **13**.

[0067] By providing the water-repellent layer **18**, the cathode electrode **15** can efficiently permeate oxygen while preventing adhesion of the organic wastewater L.

[0068] The water-repellent layer **18** is preferably made of beeswax.

[0069] Since the water-repellent layer **18** is made of beeswax, the environmental load can be reduced while maintaining the water repellency and the oxygen permeability.

[0070] The filter layer **19** is disposed on the ion exchange surface **15a** of the cathode electrode **15**. The filter layer **19** is disposed between the PEM **13** and the cathode electrode **15**. The filter layer **19** prevents the microorganism from entering the cathode electrode **15**. The filter layer **19** may or may not be disposed on the ion exchange surface **15a** of the cathode electrode **15**. The filter layer **19** is appropriately disposed.

[0071] FIG. **6** shows an example of the configuration of the cathode electrode **15**. FIG. **6** schematically shows an enlarged configuration of a part of the cathode electrode **15**. FIG. **6** shows the cathode electrode **15** formed of the non-conductive fibers **151**, the carbon-based materials **153**, and the fixing materials **155**. The cathode electrode **15** may contain a material other than the non-conductive fibers **151**, the carbon-based material **153**, and the fixing materials **155**. The cathode electrode **15** corresponds to an example of a electrodes for MFCs.

[0072] The non-conductive fibers **151** are made of a non-conductive material. The non-conductive fibers **151** support the carbon-based materials **153**. Since the cathode electrode **15** includes the non-conductive fibers **151**, the number of holes in the cathode electrode **15** increases. As the number of holes increases, a surface area of the cathode electrode **15** increases. When the surface area of the cathode electrode **15** increases, the number of sites where an oxidation-reduction reaction using oxygen occurs increases. The non-conductive fiber **151** is, for example, a defibrated fiber defibrated by a dry fiber technology. The non-conductive fiber **151** has a fiber diameter of 5 μm to 100 μm and a fiber length of 10 μm to 1000 μm . A plurality of non-conductive fibers **151** are bonded to each other by the fixing material **155**. The non-conductive fiber **151** may be made of the

same material as the fiber body **111** or may be made of a different material. The non-conductive fibers **151** are preferably made of the same material as the fiber body **111**. The non-conductive fiber **151** can be made of a common material with the fiber body **111**. The non-conductive fiber **151** corresponds to an example of a fiber and a fiber base material.

[0073] The carbon-based material **153** has conductivity. The carbon-based material **153** contributes to the conductivity of the cathode electrode **15**. The carbon-based material **153** is fixed to a surface of the non-conductive fiber **151** by the fixing material **155**. A plurality of carbon-based materials **153** are in contact with each other and fixed on the non-conductive fibers **151**. The plurality of carbon-based materials **153** are disposed in contact with each other, thereby imparting the conductivity to the non-conductive fibers **151**. The carbon-based material **153** is disposed in a single layer or a multiple layer on the non-conductive fibers **151**. The carbon-based material **153** is preferably disposed in a multiple layer on the non-conductive fibers **151**.

[0074] The carbon-based material **153** is formed of a carbon powder such as graphite particles or carbon black particles. The carbon-based material **153** is a powder having an average particle diameter of 0.1 μm to 100 μm on a volume basis. The carbon-based material **153** is formed of one or a plurality of types of carbon powder. The material of the carbon-based material **153** is not limited as long as the carbon-based material **153** has a property of imparting the conductivity to the non-conductive fibers **151**. The carbon-based material **153** corresponds to an example of a conductive material and a conductivity imparting material.

[0075] The carbon-based material **153** may be formed of a carbon fiber. The carbon fiber is produced by cutting the PAN-based carbon fiber and the pitch-based carbon fiber. The size of the carbon fibers is an average fiber diameter of 5 μm to 30 μm and an average fiber length of 0.05 mm to 0.1 mm. The fiber diameter and the fiber length of the carbon-based material **153** are smaller than the fiber diameter and the fiber length of the non-conductive fiber **151**. The carbon-based material **153** may be made of one or a plurality of types of carbon fibers. The carbon-based material **153** may be a mixture of the carbon powder and the carbon fiber.

[0076] The carbon-based material **153** may be the same material as the conductive powder **113a**, or may be a different material. The carbon-based material **153** may be the same material as the conductive fiber **113b**, or may be a different material. The carbon-based material **153** is preferably the same material as the conductive powder **113a** or the conductive fiber **113b**. Since the carbon-based material **153** is the same material as the conductive powder **113a** or the conductive fiber **113b**, the anode electrode **11** and the cathode electrode **15** can have the same basic material configuration.

[0077] The content of the carbon-based material **153** is preferably 50 mass % or more and 80 mass % or less in terms of the mass ratio of the cathode electrode **15**. When the content of the carbon-based material **153** is less than 50 mass %, the electric resistance of the cathode electrode **15** increases, and the function as the electrode deteriorates. When the content of the carbon-based material **153** is more than 80 mass %, holes in the cathode electrode **15** are reduced, and the oxygen permeability is reduced.

[0078] The fixing material **155** fixes the carbon-based material **153** to the non-conductive fibers **151**. The carbon-based material **153** is fixed to the surface of the non-conductive fiber **151** by the fixing material **155**. The fixing material **155** fixes the plurality of non-conductive fibers **151** to each other. The plurality of non-conductive fibers **151** are fixed to each other by the fixing material **155**. The fixing material **155** fixes the plurality of carbon-based materials **153** to each other. Due to the fixing material **155**, the plurality of carbon-based materials **153** are formed in one or more layers on the non-conductive fibers **151**.

[0079] The material of the fixing material **155** is not limited as long as the fixing material **155** can fix the non-conductive fibers **151** and the carbon-based material **153**. The fixing material **155** may be a solid material or a liquid material at a room temperature. The fixing material **155** preferably contains at least one of calcium carbonate, a lignin powder (average particle diameter of 26 μm to

70 μm on a volume basis), and a starch powder (average particle diameter of 1 μm to 30 μm on a volume basis). When the fixing material **155** contains at least one of the calcium carbonate, the lignin powder, and the starch powder, an environmental load when the cathode electrode **15** is generated is reduced. The fixing material **155** may be a waxy substance produced from resource organisms such as scale insects. The fixing material **155** corresponds to an example of the binder and a fixing agent.

[0080] The fixing material **155** may be the same material as the binding material **115**, or may be a different material. The fixing material **155** is preferably the same material as the binding material **115**. Since the fixing material **155** is the same material as the binding material **115**, the environmental load at the time of producing the cathode electrode **15** is reduced.

[0081] The content of the fixing material **155** is preferably 5 mass % or more and 30 mass % or less in terms of the mass ratio of the cathode electrode **15**. When the content of the fixing material **155** is less than 5 mass %, a fixing force of the carbon-based material **153** fixed on the non-conductive fibers **151** decreases. It is difficult to maintain the shape of the cathode electrode **15**. When the content of the fixing material **155** is more than 30 mass %, the fixing material **155** disposed between the plurality of carbon-based materials **153** increases, and the electric resistance increases. Electrical properties of the cathode electrode **15** deteriorate.

[0082] The cathode electrode **15** preferably contains the non-conductive fibers **151**, the carbon-based materials **153** having conductivity, and the fixing materials **155** for fixing the carbon-based materials **153** to the non-conductive fibers **151**.

[0083] Since the cathode electrode **15** includes the non-conductive fibers **151**, the carbon-based materials **153**, and the fixing materials **155**, the environmental load at the time of producing the cathode electrode **15** is reduced.

[0084] The content of the non-conductive fibers **151** in the cathode electrode **15** is a remaining amount excluding the carbon-based materials **153** and the fixing materials **155**. The cathode electrode **15** may contain an additive other than the non-conductive fibers **151**, the carbon-based materials **153**, and the fixing materials **155**. When the cathode electrode **15** contains an additive other than the non-conductive fibers **151**, the carbon-based materials **153**, and the fixing materials **155**, the content of the non-conductive fibers **151** in the cathode electrode **15** is the remaining amount excluding the carbon-based materials **153**, the fixing materials **155**, and the additive.

[0085] The cathode electrode **15** is formed using the non-conductive fibers **151**, the carbon-based materials **153**, and the fixing materials **155**. The producer mixes the non-conductive fibers **151** and the fixing materials **155** at a predetermined mass ratio to produce a mixture. In the mixture, the fixing materials **155** adhere to the surface of the non-conductive fibers **151**. After producing the mixture, the producer adds the carbon-based material **153** to the mixture and mixes them again. The producer generates a cathode electrode mixture including the non-conductive fibers **151**, the carbon-based materials **153**, and the fixing materials **155**. The producer applies heat and pressure to the cathode electrode mixture in a predetermined mold to produce a solid body. The applied pressure is in a range of 1 kN/mm² or more and 20 kN/mm² or less. A heating temperature is in a range of 80° C. or higher and 150° C. or lower. The applied pressure and the heating temperature are appropriately adjusted according to addition amounts of the carbon-based material **153** and the fixing material **155**. The producer processes the solid body into a desired shape to produce the cathode electrode **15**. The cathode electrode **15** is a compact including holes.

[0086] The MFC **10**, which comes into contact with the organic wastewater **L** containing organic substances and microorganisms, includes the anode electrode **11** including the non-conductive fiber body **111**, the conductive material **113** having conductivity, and the binding material **115** for binding the conductive material **113** to the fiber body **111**, and the cathode electrode **15** having conductivity.

[0087] Since the anode electrode **11** includes the fiber bodies **111**, the conductive materials **113**, and the binding materials **115**, the number of sites where electrons are moved by the

microorganism increases. The performance of the MFC **10** is improved.

[0088] The MFC system **100** includes: the MFC **10** having the anode electrode **11** in contact with the organic wastewater L containing an organic substance and a microorganism, the cathode electrode **15** to which oxygen is supplied, and the PEM **13** disposed between the anode electrode **11** and the cathode electrode **15**, and the resistor **31** coupled to the anode electrode **11** and the cathode electrode **15**. The anode electrode **11** includes the non-conductive fiber bodies **111**, the conductive materials **113** having conductivity, and the binding materials **115** that bonds the conductive materials **113** to the fiber bodies **111**.

[0089] Decomposition of the organic substance by the microorganism is promoted. Purification capacity of the organic wastewater L by the MFC system **100** is improved.

Example 1

[0090] In Example 1, the MFC **10** including the anode electrode **11** including the fiber body **111**, the conductive material **113**, and the binding material **115** is produced. The MFC **10** produced in Example 1 is referred to as a cell **1**.

[0091] The anode electrode **11** is formed using the fiber body **111**, the conductive material **113**, and the binding material **115**. First, a mixture containing the fiber body **111** and the binding material **115** is prepared. For the fiber body **111**, a vegetable fiber made of *eucalyptus* plant is used. For the binding material **115**, a scale insect secretion is used. The scale insect secretion is a waxy substance. A mixing ratio of the plant fiber is 80 mass %, and a mixing ratio of the scale insect secretion is 20 mass %. A planetary ball mill is used to mix the plant fiber and the scale insect secretion to prepare a mixture.

[0092] After the mixture is prepared, the conductive material **113** is mixed with the mixture. The conductive material **113** is a chopped carbon fiber, K223HM, manufactured by Mitsubishi Chemical Corporation. The chopped carbon fiber is an example of the carbon fiber. The mixing ratio of the mixture is 35 mass %, and the mixing ratio of the chopped carbon fiber is 65 mass %. The chopped carbon fiber is a short fiber having a fiber diameter of 11 μm and a fiber length of 0.05 mm. The mixture and the chopped carbon fiber are mixed by a planetary ball mill to produce an anode electrode mixture. The anode electrode mixture is a mixture of 28 mass % of the plant fiber, 7 mass % of the scale insect secretion, and 65 mass % of the chopped carbon fiber.

[0093] After the anode electrode mixture is prepared, the anode electrode mixture is heated and pressurized to prepare a sheet material. The anode electrode mixture is heated to 120° C. by a heater. The anode electrode mixture is heated and then pressurized by a pressing machine at a pressure of 5 kN/mm². The anode electrode mixture is formed into an electrode sheet having a thickness of 0.8 mm.

[0094] The electrode sheet is processed into a predetermined size after being formed. The electrode sheet is processed into a rectangular piece having a width of 20 mm and a length of 30 mm. A plurality of rectangular pieces are bonded to form the anode electrode **11** having a quadrangular prism shape. The formed anode electrode **11** is referred to as an electrode material A1.

[0095] An ion exchange resin is used for the PEM **13**. As the ion exchange resin, NAFION manufactured by DuPont is used. A NAFION membrane having a thickness of 0.5 mm is disposed on an inner peripheral surface of the anode electrode **11**. The PEM **13** of the NAFION membrane is referred to as an intermediate layer B1.

[0096] The cathode electrode **15** is made of a graphite felt manufactured by Nippon Carbon Co., Ltd. The graphite felt is a carbon-based fiber. The graphite felt is disposed on the inner peripheral surface of the PEM **13**. A PTFE membrane is disposed as the water-repellent layer **18** on the inner peripheral surface of the graphite felt. The PTFE membrane is formed by immersing one surface of the graphite felt in a PTFE solution. The cathode electrode **15** of the graphite felt is represented as an electrode material C1.

[0097] The cell **1** is produced by combining the electrode material A1, the intermediate layer B1, and the electrode material C1. The cell **1** is disposed in the organic wastewater L. The organic

wastewater L is lake water collected from the shore of Lake Suwa. The organic wastewater L contains organic mud and electrochemically active bacteria. The electrochemically active bacteria are bacteria of the genus *Geobacter*.

Example 2

[0098] In Example 2, the MFC **10** including the anode electrode **11** including the fiber body **111**, the conductive material **113**, the zero-valent iron particles **117**, and the binding material **115** is produced. The MFC **10** produced in Example 2 is referred to as a cell **2**.

[0099] The anode electrode **11** is formed using the fiber body **111**, the conductive material **113**, the zero-valent iron particles **117**, and the binding material **115**. First, a mixture containing the fiber body **111** and the binding material **115** is prepared. For the fiber body **111**, a vegetable fiber made of *eucalyptus* plant is used. The binding material **115** is a lignin powder called Sunex P321 manufactured by Nippon Paper Industries Co., Ltd. Sunex is a registered trademark. The mixing ratio of the plant fiber is 75 mass %, and the mixing ratio of the lignin powder is 25 mass %. A planetary ball mill is used to mix the plant fiber and the lignin powder to prepare a mixture.

[0100] After the mixture is prepared, the conductive material **113** and the zero-valent iron particles **117** are mixed with the mixture. The conductive material **113** is made of graphite particles manufactured by Nishimura Graphite Co., Ltd. The graphite particles are an example of the carbon powder. The graphite powder is a scaly powder having an average particle diameter of 500 μm on a volume basis. The zero-valent iron particles **117** are made of an iron powder manufactured by Fujifilm Wako Pure Chemical Industries, Ltd. The iron powder has an average particle diameter of 40 μm or less on a volume basis. The mixing ratio of the mixture is 30 mass %, the mixing ratio of the graphite powder is 65 mass %, and the mixing ratio of the iron powder is 5 mass %. The mixture, the graphite particles, and the iron powder are mixed by a planetary ball mill to prepare an anode electrode mixture. The anode electrode mixture is a mixture of 22.5 mass % of the plant fiber, 7.5 mass % of the lignin powder, 65.0 mass % of the graphite particles, and 5.0 mass % of the iron powder.

[0101] The anode electrode mixture is produced and then heated and pressurized to produce a sheet material. The anode electrode mixture is heated to 130° C. by a heater. The anode electrode mixture is heated and then pressurized by a pressing machine at a pressure of 10 kN/mm². The anode electrode mixture is formed into an electrode sheet having a thickness of 1.0 mm.

[0102] The electrode sheet is processed into a predetermined size after being formed. The electrode sheet is processed into a rectangular piece having a width of 20 mm and a length of 30 mm. A plurality of rectangular pieces are bonded to form the anode electrode **11** having a quadrangular prism shape. The formed anode electrode **11** is referred to as an electrode material A2.

[0103] The cell **2** is produced using the electrode material A2. The intermediate layer B1 is used for the PEM **13**. The electrode material C1 is used for the cathode electrode **15**. The cell **2** is produced by combining the electrode material A2, the intermediate layer B1, and the electrode material C1. The cell **2** is disposed in the organic wastewater L. The organic wastewater L is lake water collected from the shore of Lake Suwa. The organic wastewater L contains organic mud and electrochemically active bacteria. The electrochemically active bacteria are bacteria of the genus *Geobacter*.

Example 3

[0104] In Example 3, the MFC **10** including the anode electrode **11** including the fiber body **111**, the conductive material **113**, and the binding material **115** is produced. The MFC **10** produced in Example 3 is referred to as a cell **3**.

[0105] The electrode material A1 is used for the anode electrode **11**. The intermediate layer B1 is used for the PEM **13**. The cathode electrode **15** is formed using the non-conductive fibers **151**, the carbon-based materials **153**, and the fixing materials **155**. The cathode electrode **15** includes the water-repellent layer **18** made of beeswax.

[0106] The cathode electrode **15** is formed using the non-conductive fibers **151**, the carbon-based

materials **153**, and the fixing materials **155**. First, a mixture containing the non-conductive fibers **151** and the carbon-based materials **153** is prepared. For the non-conductive fibers **151**, a cellulose fiber made of recycled paper is used. The fixing material **155** is a lignin powder called Sunex P321 manufactured by Nippon Paper Industries Co., Ltd. The mixture ratio of the cellulose fiber is 75 mass %, and the mixture ratio of the lignin powder is 25 mass %. A planetary ball mill is used to mix the cellulose fiber and the lignin powder to prepare a mixture.

[0107] After the mixture is prepared, the carbon-based materials **153** are mixed into the mixture. The carbon-based material **153** are graphite particles manufactured by Nishimura Graphite Co., Ltd. The graphite particles are an example of the carbon powder. The graphite powder is a scaly powder having an average particle diameter of 500 μm on a volume basis. The mixing ratio of the mixture is 30 mass %, and the mixture ratio of the graphite powder is 70 mass %. The mixture and the graphite particles are mixed by a planetary ball mill to prepare a cathode electrode mixture. The cathode electrode mixture is a mixture of 22.5 mass % of the cellulose fiber, 7.5 mass % of the lignin powder, and 70.0 mass % of the graphite particles.

[0108] The cathode electrode mixture is produced and then heated and pressurized to produce a sheet material. The cathode electrode mixture is heated to 130° C. by a heater. The cathode electrode mixture is heated and then pressurized by a pressing machine at a pressure of 10 kN/mm². The cathode electrode mixture is formed into an electrode sheet having a thickness of 1.0 mm.

[0109] The electrode sheet is processed into a predetermined size after being formed. The electrode sheet is processed into a rectangular piece having a width of 20 mm and a length of 30 mm. A plurality of rectangular pieces are bonded to form the cathode electrode **15** having a quadrangular prism shape.

[0110] The water-repellent layer **18** provided on the cathode electrode **15** is made of beeswax manufactured by Miki Chemical Industry Co., Ltd. The beeswax is applied to the oxygen-permeable surface **15b** of the cathode electrode **15**. The applied beeswax layer functions as the water-repellent layer **18**. The cathode electrode **15** with the beeswax layer is represented as an electrode material C2.

[0111] The cell **3** is produced by combining the electrode material A1, the intermediate layer B1, and the electrode material C2. The cell **3** is disposed in the organic wastewater L. The organic wastewater L is lake water collected from the shore of Lake Suwa. The organic wastewater L contains organic mud and electrochemically active bacteria. The electrochemically active bacteria are bacteria of the genus *Geobacter*.

Comparative Example 1

[0112] In Comparative Example 1, the MFC **10** including the anode electrode **11** made of carbon cloth, which is a carbon fiber, and the cathode electrode **15** made of the graphite felt is produced. The MFC **10** produced in Comparative Example 1 is referred to as a cell **4**.

[0113] The anode electrode **11** is made of carbon cloth manufactured by Toray Industries, Inc. The thickness of the carbon cloth is 100 μm . The anode electrode **11** using the carbon cloth is referred to as an electrode material A3.

[0114] The cell **4** is produced by combining the electrode material A3, the intermediate layer B1, and the electrode material C1. The cell **4** is disposed in the organic wastewater L. The organic wastewater L is lake water collected from the shore of Lake Suwa. The organic wastewater L contains organic mud and electrochemically active bacteria. The electrochemically active bacteria are bacteria of the genus *Geobacter*.

[0115] FIG. 7 shows the configuration of each MFC **10**. FIG. 7 shows the configurations of the cell **1**, the cell **2**, the cell **3**, and the cell **4**, which are the MFCs **10**. The cell **1**, the cell **2**, and the cell **3** are the configurations of the MFCs **10** produced in Examples 1, 2, and 3, respectively. The cell **4** has the configuration of the MFC **10** produced in Comparative Example 1.

Evaluation Result

[0116] FIG. 8 shows an example of evaluation results of the MFC 10. FIG. 8 shows a temporal change in generated voltages of the MFC 10. FIG. 8 shows the temporal changes in the generated voltages of the cells 1, 2, 3, and 4. A horizontal axis represents elapse of days. A vertical axis represents the generated voltage.

[0117] As shown in FIG. 8, the generated voltages of the cell 1, the cell 3, and the cell 4 at an initial stage of the beginning of measurement are substantially equal values. With the elapse of days, the generated voltage of the cell 4, which is the comparative example, decreases. In the cell 1, the cell 2, and the cell 3 according to the embodiment, the decrease in the generated voltage with the lapse of days is prevented. The generated voltage remains at a constant value.

[0118] The generated voltage by the cell 2 including the zero-valent iron particles 117 is higher than the generated voltages by the cells 1, 3, and 4. In the cell 2, a decrease in the generated voltage due to the lapse of days is prevented. The inclusion of the zero-valent iron particles 117 improves cell characteristics.

[0119] The generated voltage by the cell 3 having the water-repellent layer 118 made of beeswax remains higher than the generated voltage by the cell 1 having the water-repellent layer 118 made of PTFE after the elapse of a predetermined number of days. The use of beeswax for the water-repellent layer 118 improves the cell characteristics.

[0120] FIGS. 9, 10, 11, and 12 show examples of the evaluation results of the MFC 10. FIGS. 9, 10, 11, and 12 show polarization analysis results of the MFC 10. FIGS. 9, 10, 11, and 12 show current-voltage characteristics of the MFC 10 and a relationship between a current and electric power. FIGS. 9, 10, 11, and 12 show the current-voltage characteristics and the relationship between the current and the electric power after the elapse of 12 days from the start of evaluation of the MFC 10. Current and voltage measurement is performed by changing a resistance value of the resistor 31 coupled to the MFC 10.

[0121] FIG. 9 shows the current-voltage characteristics of the cell 1 and the relationship between the current and the electric power. FIG. 10 shows the current-voltage characteristics of the cell 2 and the relationship between the current and the electric power. FIG. 11 shows the current-voltage characteristics of the cell 3 and the relationship between the current and the electric power. FIG. 12 shows the current-voltage characteristics of the cell 4 and the relationship between the current and the electric power.

[0122] As shown in FIGS. 9, 10, 11, and 12, a maximum power value, which is a peak value of the electric power, differs depending on the cell. The maximum power values of the cells 1, 2, and 3 are higher than the maximum power value of the cell 4. The use of the anode electrode 11 including the fiber body 111, the conductive material 113, and the binding material 115 improves the characteristics of the MFC 10.

[0123] The maximum power value of the cell 2 is higher than the maximum power values of the cell 1 and the cell 3. The addition of the zero-valent iron particles 117 to the anode electrode 11 improves the characteristics of the MFC 10. The maximum power value of the cell 3 is higher than the maximum power value of the cell 4. The use of beeswax in the water-repellent layer 18 improves the characteristics of the MFC 10.

Claims

1. A electrodes for MFCs comprising: a non-conductive fiber; a conductive material having conductivity; and a binder configured to bind the conductive material to the fiber.
2. The electrodes for MFCs according to claim 1, wherein the conductive material is at least one of a carbon fiber and graphite particles.
3. The electrodes for MFCs according to claim 2, further comprising: metal particles having conductivity.
4. The electrodes for MFCs according to claim 1, wherein the binder contains at least one of

calcium carbonate, a lignin powder, and a starch powder.

5. A MFC to be in contact with a liquid containing an organic substance and an electrochemically active bacterium, the MFC comprising: an anode electrode containing a non-conductive fiber, a conductive material having conductivity, and a binder for binding the conductive material to the fiber; and a cathode electrode having conductivity.

6. The MFC according to claim 5, wherein the cathode electrode contains a non-conductive fiber base material, a conductivity imparting material having conductivity, and a fixing agent for fixing the conductivity imparting material to the fiber base material.

7. The MFC according to claim 6, wherein the cathode electrode includes a water-repellent layer.

8. The MFC according to claim 7, wherein the water-repellent layer is made of beeswax.

9. The MFC according to claim 5, wherein the anode electrode contains metal particles having conductivity.

10. The MFC according to claim 5, further comprising: an PEM containing a non-conductive fiber material, wood powder particles, and an adhesive for bonding the fiber material to the wood powder particles, and provided between the cathode electrode and the anode electrode in contact with the cathode electrode and the anode electrode.

11. A MFC system comprising: a MFC including an anode electrode to be in contact with a liquid containing an organic substance and an electrochemically active bacterium, a cathode electrode to which oxygen is supplied, and an PEM disposed between the anode electrode and the cathode electrode; and a resistor coupled to the anode electrode and the cathode electrode, wherein the anode electrode contains a non-conductive fiber, a conductive material having conductivity, and a binder configured to bind the conductive material to the fiber.
